Textual Risk Disclosures and Investors' Risk Perceptions

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Abstract
We examine the association between changes in companies' textual risk disclosures in 10-K filings and changes in stock market and analyst activity around the filings. We find that annual increases in risk disclosures are associated with increased stock return volatility and trading volume around and after the filings. Increases in risk disclosures are also associated with more dispersed forecast revisions around the filings. In contrast to prior literature documenting resolved uncertainties in response to various types of company disclosures, our findings suggest that textual risk disclosures increase investors' risk perceptions. However, the results are less pronounced for firm-level disclosures that deviate from those of other companies in the same industry and year. These results lend support for critics' arguments that firm-level risk disclosures are more likely to be boilerplate.

JEL Classification: D8, G24, G12, M4.

Keywords: Disclosure, risk, uncertainty, 10-K filings, trading volume, stock return volatility, analysts.

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"To know what we know, and know what we do not know, is wisdom" Confucius

1 Introduction

A long-standing criticism of financial reporting is the lack of useful disclosures about company risks and uncertainties (AICPA 1987; Schrand and Elliott 1998). This criticism has become more important amidst large market-wide fluctuations in the last decade (Kaplan 2011). Regulators have traditionally responded to market-wide fluctuations by encouraging corporations to make more meaningful risk disclosures (Jorgensen and Kirschenheiter 2003).

In this study, we investigate the informativeness of textual risk disclosures in company annual reports filed with the SEC between years 1994 and 2007. Textual risk disclosures, which have grown in length and content during the sample period, present users with managers' assessments about future contingencies and a range of exposures to market factors. Textual risk disclosures differ from other corporate disclosures in that they guide users about the *range* of future performance rather than the *level* of future performance. This distinction is reflected in how we test the informativeness of textual risk disclosures. We hypothesize that informative risk disclosures will change users' risk perceptions, i.e., the range of users' predictions of future performance as well as users' confidence in their predictions.

We test three competing arguments regarding how risk disclosures affect users' risk perceptions. The first argument is that risk disclosures are boilerplate (hereafter, the *null argument*). The second argument is that risk disclosures reveal unknown risk factors and contingencies, thereby increasing users' risk perceptions (hereafter, the *divergence argument*). The third argument is that risk disclosures resolve a company's known risk factors and contingencies, thereby reducing users' risk perceptions (hereafter, the *convergence argument*). Companies are likely to repeat a significant portion of their risk disclosures over consecutive

annual reports. In an effort to address concerns related to correlated omitted variables and reverse causality, we employ a changes analysis and investigate how annual changes in risk disclosures change users' risk perceptions, as measured by stock return volatility, trading volume, and analysts' forecast revisions around the filing dates. Our empirical methodology heeds Li's (2010a) call for using a change specification whenever appropriate in textual analysis research in order to mitigate endogeneity concerns.

Our findings support the divergence argument. The annual increase in the number of risk sentences in a company's 10-K filing is associated with higher return volatility (particularly higher volatility of negative returns) and higher trading volume during the 60 trading-day period after the filing relative to the 60 trading-day period before the filing; a higher three-day trading volume around the filing; and more volatile analyst forecast revisions surrounding the filing. Our results are robust to controls for other information in the 10-K filings, changes in length and complexity of the filings, changes in performance, ownership, managerial earnings forecasts, and changes in market-level economic factors around the filings. The effect of risk disclosures is economically significant relative to the effect of the market-level control variables in our models. Our finding of higher dispersion in forecast revisions differs from literature that usually documents reduced forecast dispersions after corporate disclosures (Lang and Lundholm 1996; Nichols and Wieland 2009). This discrepancy is not entirely surprising; risk disclosures inform investors about contingencies and risk factors that were unknown to investors.

An important question unanswered in the above findings is whether idiosyncratic risk disclosures are more informative than industry-wide risk disclosures. To investigate, we reperform our analyses after dividing the change in a company's number of risk sentences into two components: i) median change in the number of risk sentences of other companies in the

same industry and fiscal year, and ii) the deviation from (i). In general, we observe stronger relations between industry-level risk disclosures and changes in users' risk perceptions, suggesting that firm-specific disclosures are less informative than industry-level disclosures.

Our study contributes to the risk disclosure literature. Prior literature examines the effect of SFAS 119 derivative disclosures and Financial Reporting Release (FRR) No. 48, which requires companies to disclose (largely in tabular format) exposures of financial assets and liabilities to market factors such as interest rates, exchange rates, and commodity prices (Rajgopal 1999; Wong 2000; Jorion 2002; Linsmeier et al. 2002). While this literature converges that FRR No. 48 and SFAS 119 disclosures are informative, it is unclear from these studies whether and how textual risk disclosures are informative for several reasons. First, FRR No. 48 and SFAS 119 mandate companies to disclose specific quantitative information about the known exposures to market factors. Therefore, this prior evidence hinges on a setting where investors' risk perceptions are bound to converge with additional disclosures. Second, textual risk disclosures cover a broader spectrum of risk factors, such as operational and legal risks, which the prior literature does not examine. These types of risk are also more difficult to assess than SFAS 119 and FRR No. 48 disclosures, which deceases the generalizability of the prior findings to the broader risk disclosure setting. Third, the empirical analyses in prior literature predate the two major economic crises in the recent decade (i.e., 2000 and 2008), raising additional concerns about generalizing the prior literature's conclusions to more recent periods.

Our study also contributes to the textual analysis literature. Li (2006) expands the scope of risk disclosures using a similar method with ours and finds that companies signal bad future earnings through textual risk disclosures and that stock returns of these companies underperform after the filings, consistent with investors underreacting (or not reacting) to these signals. We

extend Li (2006) by documenting that users react as if textual risk disclosures inform them about unknown risk factors. In a contemporary paper, Campbell et al. (2011) find that the length of Section 1A in 10-K filings, which is mandated in 2005 as a narrative outlet for company risk factors, reduces information asymmetry, which is proxied by bid-ask spreads, but increases investors' risk perceptions, which is proxied by beta and stock return volatility. There are significant differences between our paper and Campbell et al. (2011) with respect to sample characteristics (e.g., our sample years of 1994 to 2007 versus Campbell et al.'s 2005 to 2009) as well as research design choices (e.g., we use a changes methodology while Campbell et al. use levels of risk disclosures as the key independent variable), preventing a direct comparison of the findings. Yet, both papers converge that risk disclosures are informative. Lehavy et al. (2011) find that readability of 10-K filings affects analyst forecast dispersion, accuracy, and effort. Similarly, You and Zhang (2009) find that investors underreact to longer 10-K filings, pointing to the time and effort spent on interpreting the filings. We find that risk disclosures have an incremental effect on investors' and analysts' risk perceptions over the effect of the readability and complexity measures.

The remainder of the paper is organized as follows. Section 2 provides hypothesis development in light of previous theoretical and empirical research. Section 3 describes the sample selection and research design. Section 4 presents empirical results. Section 5 concludes.

2 Hypothesis Development

A large body of research finds that forward-looking disclosures, such as those in 10-K filings, managerial forecasts, press releases, and conference calls, resolve corporate uncertainties

(e.g., Clement et al. 2003; Mohanram and Sunder 2006; Beyer et al. 2010; Muslu et al. 2011).[1] Though intrinsically forward-looking, risk disclosures differ from forward-looking disclosures in that they explain but do not necessarily resolve corporate uncertainties. That is, rather than informing users about a point forecast of performance that users can converge around, risk disclosures provide information about the second moment of expected performance. Hence, risk disclosures have the potential to increase as much as to decrease users' risk perceptions (Kim and Verrechia 1994; Cready 2007).

2.1 Regulatory environment for corporate risk reporting

The primary objective of financial reporting is to provide useful information to assess the amount, timing, and uncertainty of future net cash inflows to the entity (FASB 2010). Several standards require or encourage companies to disclose uncertainties. SFAS No. 106 requires disclosures about potential changes in postretirement benefit plan costs (FASB 1990). SFAS No. 133, which superseded SFAS No. 119 and was later amended by SFAS 155, encourages companies to disclose quantitative information about market risks of derivative instruments and hedging activities (FASB 1998). SFAS No. 140 requires that companies with securitized financial assets disclose information about key assumptions made in determining fair values of retained interests (FASB 2000). The Private Securities Litigation Reform Act of 1995 establishes a safe harbor from liability in private lawsuits for companies making meaningful risk statements that accompany forward-looking statements.

Corporate risk reporting receives particular regulatory attention after market downturns and volatilities. For instance, corporate losses from financial transactions in the early 1990s

[1] Not all forward-looking disclosures resolve uncertainties. Rogers et al. (2009) document higher implied volatilities derived from exchange-traded options around managerial forecasts (especially around irregular managerial forecasts and forecasts that convey bad news).

prompted calls for expanded disclosures on financial instruments (Linsmeier and Pearson 1997). In January 1997, the SEC issued FRR No. 48 requiring firms to provide information about market risk factors related to their trading and non-trading instruments, such as those related to stock prices, interest rates, exchange rates, and commodity prices (SEC 1997).[2] Similarly, after stock market declines from 2000 to 2002, the SEC mandated that companies discuss risk factors in the first pages (Section 1A) of 10-K filings.[3] In its interpretive guidance, the SEC states "…in identifying, discussing, and analyzing known material trends and uncertainties, companies are expected to consider all relevant information, even if that information is not required to be disclosed" (SEC 2003). The economic crisis of 2008 resulted in more regulatory oversight on risk disclosures. The SEC has intensified review of risk disclosures in corporate filings and used comment letters to require more risk information from specific companies and industries (Johnson 2010). The Dodd-Frank Act of 2010 creates regulatory agencies that are mandated to search for unforeseen risks in the financial system (Financial Stability Oversight Council and Office of Financial Research), and grants the SEC and the Federal Reserve more authority to improve transparency in the financial system and corporate governance.

2.2 Challenges in reporting corporate risk

Financial authorities require companies to make 'meaningful' risk disclosures. This is evidenced in SEC's intensified requests for clarification from companies believed to have used boilerplate statements and courts' ruling that fixed and cryptic cautionary language does not satisfy the safe harbor provision of the Private Securities Litigation Reform Act (Nelson and Pritchard 2007). Despite the regulatory environment, companies may easily avoid providing

[2] FRR No. 48 mandates these disclosures to be made as Item 7A as described in Item 305 of Regulation S-K introduced under the Securities Exchange Act of 1934, which had encouraged registrants to provide market risk disclosures.

[3] These factors have to be provided under the caption "Risk Factors" (as Item 1A in the 10-K filing) as described in Item 503(c) of Regulation S-K introduced under the Securities Exchange Act of 1934.

useful risk-related information. For instance, a statement like "*Our company may not be able to implement its growth strategy*" may help companies comply with regulations, however, unless accompanied by specific details it is likely not informative to users in assessing corporate risks. Several factors contribute to this deficiency. First, corporate risk assessments are often regarded as negative information (Li 2006), which managers tend to withhold because of career concerns (Kothari et al. 2009). Second risk assessments include proprietary information, which companies tend to withhold to reduce competition (Dye 1985). Finally, given their fluid nature, a company's risk exposures are hard to perceive and measure, even by insiders. Kaplan (2011) states "How can we quantify risk or develop risk indicators for an event that has not yet occurred and, we hope, may never occur?" The general lack of corporate warnings before the near-collapse of the financial system in 2008 is recent evidence of unrecognized or mismeasured risks.

2.3 Reporting known and unknown risks

The psychology and economics literature have long distinguished between known and unknown risks. Knight (1921) defines 'risk' as decision situations with available probabilities to guide choice, and 'uncertainty' as decision situations in which information is too imprecise to be summarized by probabilities. Similarly, Slovic et al. (1980) define known risk as probabilities of future outcomes that can be perceived by individuals, and unknown risk as unobservable or uncontrollable future outcomes that adversely affect individuals' judgments. Investors' distaste for unknown risks, also known as ambiguity aversion or information risk, affect asset prices over and above the effect of traditional risk factors (Barry and Brown 1985; Epstein and Schneider 2008; Caskey 2009).[4] In an experimental setting, Koonce et al. (2005) find that investors' risk

[4] A related strand of literature examines how information precision and information asymmetry affects the cost of capital (Diamond and Verrecchia 1991; Botosan 1997; Francis et al. 2004; Easley and O'Hara 2004; Lambert et al. 2007; Bhattacharya et al. 2009; Kravet and Shevlin 2010).

assessments are affected by unknown and dread, the two behavioral factors of Slovic (1987), besides the conventional decision variables such as probabilities and outcomes.

Prior research on risk reporting does not distinguish between known and unknown risks, and documents that mandated disclosures provide useful information about market risk factors (Hodder et al. 2001).[5] Rajgopal (1999) finds that oil and gas firms' disclosures about market exposures are associated with stock return sensitivities to oil and gas prices. Linsmeier et al. (2002) find that trading volume sensitivity to changes in market risk exposures declines after firms disclose information mandated by FRR No. 48. Jorion (2002) finds that banks' Value at Risk (VaR) disclosures predict trading revenues. In contrast, Wong (2000) finds only weak evidence that derivative disclosures help predict currency exposures. These studies are limited to the disclosures of known market risk factors and generally find that investors' risk perceptions decrease after these disclosures.

2.4 Information content of risk disclosures

The quantifiable market-wide risk factors comprise a small share of corporate risk factors and contingencies, which include those related to competition, suppliers, employees, customers, financing, foreign operations, regulations, litigation, governance, and environment. The severity of this disclosure gap is only addressed by the recent regulations mandating that companies discuss their quantitative and qualitative assessments about risks and uncertainties. As such, the quality of risk disclosures remains largely voluntary despite the efforts of regulators, and the informativeness of such disclosures, to our knowledge, is largely unknown.

Two recent papers address the informativeness of risk disclosures from different angles. First, Li (2006) documents that an increase in risk sentiment in annual reports (as captured by

[5] The theoretical literature on risk disclosures focuses on cost of capital. Jorgensen and Kirschenheiter (2003) propose a partial disclosure equilibrium, in which managers voluntarily disclose (not disclose) if their firm has low (high) variance of future cash flows; and the disclosing firm has a lower risk premium ex post.

count of words ‘risk’ and ‘uncertainty’) is associated with poor future stock returns, suggesting that investors underreact to the risk sentiment of annual reports. Second, Campbell et al. (2011) find that the length of Section 1A’s of the 10-K filings (in which companies identify their risk factors after December 2005) is associated with low bid-ask spreads (to proxy for information asymmetry) and high beta and stock return volatility (to proxy for investors’ assessments of fundamental risk) in the following year. We complement this evidence by investigating how changes in textual risk disclosures affect users’ risk perceptions.

Our study also contributes to the general literature studying textual information. Li (2008) finds that the readability of annual reports is associated with earnings persistence and Lehavy et al. (2011) find that the readability of annual reports affects investors’ decisions. Others examine the effect of tone in annual reports (Kothari et al. 2009; Feldman et al. 2010; Davis and Tama-Sweet 2011). More closely related to our study, Nelson and Pritchard (2007) show that firms increase their cautionary language in annual reports in response to litigation risk. Li (2010b) finds that the tone in forward-looking statements in the MD&A section is associated with future earnings where statements related to risk and uncertainty are a (negative) category of tone. We extend this literature by differentiating textual risk disclosures from the notion of tone. In addition, we analyze the effect of negative tone in risk disclosures.

2.5 Measuring risk disclosures

We develop a UNIX Perl code that, in sequence, (1) downloads 10-K filings from the SEC *Edgar* database for fiscal years between 1994 and 2007, (2) extracts textual risk disclosures from the 10-K filings, and (3) analyzes the content of these disclosures. Prior to 2002, companies filed their 10-K filings in the ASCII-code format. After 2002, companies have uploaded their annual reports in various formats, such as text, html, or pdf formats. Since the Perl code more

accurately processes ASCII-code files than other file types, we supplement our post-2002 sample using annual reports obtained from the *10-K Wizard* database.[6] We use the CIK of the Edgar filings to match observations with Compustat to then obtain the required financial data.

The Perl code parses the annual report into sentences after excluding Sections 3 and 4, which include appendices about executive biographies, third-party transactions, and legal documents. Next, the code tags a sentence as risk-related if the sentence includes at least one of the following risk-related keywords (where * implies that suffixes are allowed): can/cannot, could, may, might, risk*, uncertain*, likely to, subject to, potential*, vary*/varies, depend*, expos*, fluctuat*, possibl*, susceptible, affect, influenc*, and hedg*. The keyword list is developed based on our reading of 100 randomly selected annual reports. We define *Risk Disclosure*$_{i,t}$ as the total number of sentences with at least one risk-related keyword.[7] Given that many risk exposures will change little over time, companies are likely to repeat their risk assessments over consecutive filings. We therefore adopt a changes methodology to understand how users react to companies' new risk disclosures. In order to capture new risk disclosures in 10-K filings, our research design uses Δ*Risk Disclosure*$_{i,t}$, defined as the difference between *Risk Disclosure*$_{i,t}$ and *Risk Disclosure*$_{i,t-1}$. We do not scale this variable but instead include in our model the change in number of non-risk sentences in 10-K filings to control for the overall change in the size of the annual report.

Table A1 of Appendix A presents the average number of risk-related keywords in a 10-K filing. On average, there are 170 risk-related keywords in a 10-K filing during our sample period.

[6] We use the TextPipe Software to convert rich text formatted files from *10-K Wizard* to ASCII-code formatted files.
[7] We acknowledge that this algorithm does not perfectly measure the intensity of risk disclosures in annual reports. For instance, our algorithm defines tables as single sentences, some of which present extensive information about how future performance can vary with respect to various factors. Furthermore, we do not differentiate between audited risk statements that are in the footnotes and unaudited risk statements that are elsewhere in the annual report. However, the changes methodology of our tests should prevent such measurement errors affecting our conclusions. Further, our out-of-sample validation tests (untabulated) show that our routine is well-specified and powerful.

The risk-related keywords that contribute most to our risk disclosure measure are may, could, can/cannot, risk, subject to, and affect. Table A2 provides examples of new sentences in 10-K filings that include risk-related keywords. These examples appear to provide specific information about future risks and uncertainties that companies may face. Appendix B provides an anecdotal example of informative risk disclosures. Take-Two Interactive (TTWO) disclosed new information regarding uncertainty about employment contracts and financing constraints in its January 31, 2006 10-K filing. This filing was followed by a related news article (from Consumer Electronics Daily 2006) and a period of increased daily stock return volatility.

The literature proposes two other measures of textual risk disclosure. Li (2006) counts the number of six risk-related words ("risk", "risks", "risky", "uncertain", "uncertainty", and "uncertainties") in the annual report, and uses the annual difference of the logarithm of this count as his risk disclosure measure. Campbell et al. (2011) use the total word count in Section 1A of the annual report, in which companies are mandated since December 2005 to discuss risk factors. There are several differences between our risk disclosure measure and those in Li (2006) and Campbell et al. (2011). First, our measure is based on a count of sentences rather than words. We define a sentence as the unit of analysis instead of a word, because a sentence is the smallest integral unit of text that conveys an idea (Ivers 1991). In other words, by using sentences instead of words, we avoid multiple counting of the same risk-related information. However, word and sentence counts are highly correlated.[8] Second, in comparison to Li's (2006) measure, we develop a broader list of risk disclosure. Besides the word roots of "risk" and "uncertain", our method counts 16 other word roots that indicate risk and uncertainty of future outcomes. Third, given that the risk disclosures appear anywhere in the annual report, we search for risk-related

[8] Alternative methods to measure changes in textual risk disclosures involve examining the rate of change in the frequency of specific words within the text or frequency of word groups within a sentence (Brown and Tucker 2011; Nelson and Pritchard 2007).

keywords between Sections 1 and 14 of the annual report. This search is wider than Campbell et al. (2011), which searches Section 1A only, but narrower than Li (2006), which searches the whole annual report including the exhibits and financial schedules. Table A3 of Appendix A presents the average total number of sentences and risk-related sentences as well as the change in these sentences by the section of the annual report. We find that risk disclosures appear most frequently in Sections 1 (Business), 1A (Risk factors), 7 (MD&A), and 8 (Financial statements). We have omitted the exhibits and financial schedules in order to reduce Type II errors, i.e., detecting risk-related keywords that are not informative to the user (e.g., in legal documents).

We also calculate the correlation between our measure and these other measures based on our sample.[9] Δ*Risk Disclosure* strongly correlates with Li's (2006) measure with a correlation coefficient of 0.76. Δ*Risk Disclosure* also correlates with Campbell et al.'s (2011) measure with a correlation coefficient of 0.14. If we convert the Campbell et al. (2011) measure to "changes" instead of "levels", then the correlation coefficient increases to 0.22.

2.6 Predictions

We examine how Δ*Risk Disclosure* is associated with the range of investors' and analysts' predictions as well as their confidence levels in their predictions around the filings. Given the length of the 10-K filings, we expect that investors cannot promptly update their predictions (You and Zhang 2009). Therefore, we keep the testing period long enough (60 trading days) to allow for investors and analysts to interpret the risk disclosures and react based

[9] For this comparison, we calculate the Li (2006) and Campbell et al. (2011) risk disclosure measures on sentence basis rather than word basis, because this is the form for which we have the data to calculate these measures.

on their interpretations, but short enough to prevent the effect of confounding events, such as the disclosure of other information about corporate risk.[10]

2.6.1 Risk disclosures and stock return volatility

Morck et al. (2000) argue that increased firm-level return volatility indicates more detailed firm-specific information being incorporated into stock prices.[11] Shalen (1993) and Garfinkel (2009) use daily stock return volatility to proxy for diverging investor opinions. If risk disclosures introduce unknown contingencies and risk factors, users will diverge in their predictions of future performance and users' confidence in their predictions will decrease (divergence argument). Therefore, the divergence argument predicts higher return volatility during the first 60 trading days after the filings than the last 60 trading days before the filings, reflecting the increased range and reduced confidence levels in investors' predictions. If, on the other hand, risk disclosures resolve known contingencies and risk factors, users will converge in their predictions and increase their confidence level (convergence argument). Therefore, the convergence argument predicts lower post-filing return volatility. We test the above predictions using the change in the volatility of stock returns from the 60 trading-day period before to the 60 trading-day period after the 10-K filing, $\Delta\sigma(\textit{Return})$, multiplied by 100. We exclude from this calculation the three trading-day filing period window [-1, 1], where day 0 is the filing date.

Given that return volatility is a symmetric risk measure and that risk disclosures are criticized for lacking information about negative eventualities, our second test focuses on the volatility of negative stock returns. If risk disclosures increase (decrease) users' risk perceptions, then the effect on downside risk will increase (decrease) more relative to the effect of upside

[10] Our analysis is constrained by the possibility of other news that may correlate with risk disclosures over long windows. Therefore, our causality interpretation is potentially confounded, but—we argue—this is less likely with our study than for studies investigating changes in longer horizons such as years.

[11] There are also arguments that firm-level stock return volatility is associated with noise and less information about the company (Roll 1988), however this view seems to have lost support in recent years (Liu and Wysocki 2007).

risk, and therefore, daily negative stock returns will be more (less) volatile than daily positive stock returns. We measure this prediction using the change in the ratio of volatility of negative stock returns to volatility of positive stock returns from the 60 trading-day period before to the 60 trading-day period after the 10-K filing, Δ(σ*(Neg Return*)/σ(*Pos Return*)) (McAnally et al. 2011). σ(*Neg Return*) (σ(*Pos Return*)) is the standard deviation of daily stock returns during trading days with negative (positive) returns, where days with positive (negative) returns are valued at zero. We exclude from this calculation the three trading-day filing period window [-1, 1], where day 0 is the filing date.

2.6.2 Risk disclosures and trading volume

Garfinkel (2009) shows that unexplained trading volume is the most reliable proxy for opinion divergence. Trading volume around earnings announcements reflects individual investors' differential belief revisions (Kim and Verrecchia 1991; Karpoff 1986).[12] Bamber et al. (1999) show that trading volume around earnings announcements increases with measures of differential interpretations. If risk disclosures in 10-K filings increases (decreases) the range of investors' predictions, there will be greater (lesser) differential belief revisions and thus increased (decreased) trading volume. Accordingly, we examine the short-window trading volume around the 10-K filings. We define *Log*(*Filing Volume*) as the logarithm of average trading volume divided by outstanding shares (+0.000255 to avoid taking the log of zero) in the three-day window around the 10-K filing (Cready and Hurtt 2002).

In addition, the divergence (convergence) argument predicts that investors trade more (less) after the filings relative to before the filings if they have lower (higher) confidence in their predictions. Lower (higher) confidence makes it more (less) likely that investors change their

[12] Differential belief revision around disclosures can arise from either: (1) differential interpretations of the disclosures, or (2) differences in the precision of investors' pre-disclosure information.

expectations of firm value based on the arrival of new information and trade shares. We define $\Delta Log(Volume)$ as the change in a company's logarithm of the average trading volume divided by outstanding shares from the 60 trading-day period immediately before the 10-K filing to the 60 trading-day period immediately after the filing.

2.6.3 Risk disclosures and analysts' differential interpretations

Dispersion in analyst forecasts increases with uncertainty (Barron et al. 1998). If risk disclosures increase users' risk perceptions–especially if investors do not know about the reported risk factors–then analysts will diverge in their beliefs. This expectation is in line with Barron et al. (2002), who document that commonality of information among active analysts decreases around earnings announcements, and with Kim and Verrecchia (1991), who argue that analysts generate idiosyncratic information around earnings announcements.

On the other hand, risk disclosures can update a company's assessments of risk factors, on which analysts can converge their beliefs–especially if investors are aware of the reported risk factors. The literature generally suggests that corporate disclosures reduce the variance of expected future cash flows and thus reduce dispersion of analyst forecasts (Lambert et al. 2007). Distinct from the above arguments, the null argument predicts that analysts will not revise their forecasts differently if they assess risk disclosures to be uninformative.

We use the volatility of analysts' forecast revisions to capture analysts' differential interpretations (Lang and Lundholm 1996). $\sigma(Forecast\ Revision)$ is defined as the standard deviation of individual analysts' forecasts revisions. An analyst's forecast revision is calculated as the analyst's first forecast during the first two months after the 10-K filing minus the analyst's

last forecast during the last two months before the filing. We require at least five analysts to compute this variable.[13]

2.6.4 Firm- versus industry-level risk disclosures

Risk disclosures are primarily criticized for lack of firm-specific information (Johnson 2010). We examine the validity of this criticism by dividing the changes in risk sentences into their industry-level and firm-level components. Δ*Risk Disclosure* is divided into Δ*Industry-Level RD*, defined as the four-digit SIC industry and year median of Δ*Risk Disclosure*, and Δ*Firm-Level RD*, defined as Δ*Risk Disclosure* net of Δ*Industry-Level RD*. The changes in a company's risk disclosure that conform with (deviate from) those of the company's peers are more likely to be industry-specific (firm-specific). This is because mandated risk disclosures stemming from different channels such as new standards, the SEC's interpretations, or the SEC's comment letters affect companies in the same industry similarly in the same year.

3 Research Design

Our sample includes firm-year observations with 10-K filings on the SEC *Edgar* universe between years 1994 and 2007, at least one analyst following recorded on I/B/E/S, non-missing data from Compustat and CRSP databases, and non-missing data for the previous year. Our sample is composed of 28,110 firm-year observations from 4,315 unique firms. The number of observations decreases for the test of analyst forecast revisions because of the data requirements to calculate this variable.

We test how changes in risk disclosures relate with changes in activities of investors and analysts before and after the filings. We use a changes model in order to examine the effect of new risk disclosures and address potential correlated omitted variables. We estimate the

[13] A meaningful number of analysts is needed to compute the standard deviation of forecast revisions. The results are similar if the number of analysts used is higher or lower than five.

following OLS model (with modifications across different tests) on a pooled times-series cross-sectional basis:

$$\begin{aligned} Y = {} & \beta_0 + \beta_1 \Delta Risk\ Disclosure + \beta_2 \Delta Non\text{-}Risk\ Disclosure + \beta_3 \Delta Market\ Return\ Volatility \\ & + \beta_4 \Delta Fog\ Index + \beta_5 \Delta Institutional\ Ownership + \beta_6 \Delta Managerial\ Forecast \\ & + \beta_7 \Delta Sales\ Growth + \beta_8 \Delta ROA + \beta_9 \Delta Segments + \beta_{10} \Delta Loss + \beta_{11} Filing\ Return \\ & + \beta_{12} Absolute\ Filing\ Return + \beta_{13} \Delta Market\ Return + \varepsilon \end{aligned} \quad (1)$$

where, in separate tests, Y is equal to the following proxies about changes in users' risk perception: change in volatility of daily stock returns, $\Delta\sigma(Return)$; change in the ratio of volatility of negative stock returns to volatility of positive stock returns, $\Delta(\sigma(Neg\ Return)/\sigma(Pos\ Return))$; change in the three-day trading volume around the 10-K filing, $Log(Filing\ Volume)$; change in trading volume, $\Delta Log(Volume)$; and dispersion of forecast revisions, $\sigma(Forecast\ Revision)$.

The base model controls for changes in the overall length of the 10-K filing by including the number of non-risk sentences, $\Delta Non\text{-}Risk\ Disclosure$. We also include the change in market volatility surrounding the 10-K filing to control for market-wide economic factors. $\Delta Market\ Return\ Volatility$ is the change in market-level stock return volatility from the 60 trading-day period before to the 60 trading-day period after the filings, multiplied by 100.

We also separately estimate our model including a number of control variables related to firm characteristics and other information disclosed in the 10-K that could affect investor and analyst activity. We control for the readability of the annual report, $\Delta Fog\ Index$, because prior literature finds this attribute is associated with investors' decisions (Li 2008; You and Zhang 2009) and can potentially affect uncertainty. The quarterly change in the percentage of institutional ownership, $\Delta Institutional\ Ownership$, is included because prior research finds an association between institutional ownership and stock return volatility (Potter 1992; Bushee and

Noe 2000). We control for the change in the number of management forecasts from two months before to two months after the filings, Δ*Managerial Forecast*, because management forecasts are associated with increased uncertainty (Clement et al. 2003; Rogers et al. 2009). We control for other changes in firm operating characteristics by including quarterly change in seasonally-adjusted sales growth, Δ*Sales Growth*; the quarterly change in seasonally-adjusted net income divided by total assets, Δ*ROA*; the annual change in the number of business segments, Δ*Segments*; and an indicator variable for whether the company switched from a quarterly profit to a loss, Δ*Loss*. We control for the overall information in the 10-K filings using the signed and absolute value of company's stock returns during the three-day window around the 10-K filing, *Filing Return* and *Absolute Filing Return*, respectively. Finally, we further control for changes in market-wide economic factors using the change in the value-weighted market return from the 60 trading-day period before to the 60 trading-day period after the filings, Δ*Market Return*.[14] Appendix C defines the variables. All continuous variables are winsorized at the 1st and 99th percentile to mitigate the effect of outliers. Further, influential observations with studentized t-statistics greater than two are excluded in each test. The standard errors are adjusted for heteroskedasticity and are clustered for firm and filing month.[15]

4 Results

4.1 Descriptive statistics

Panel A of Table 1 presents descriptive statistics. The mean (median) value of Δ*Risk Disclosure* is 13.7 (6.0) over our sample period. The untabulated mean (median) of the level variable, *Risk Disclosure*, is 109.9 (87.0), suggesting that risk disclosures in annual reports grow

[14] Δ*Market Return Volatility* and Δ*Market Return* are calculated excluding daily returns from the three-day window centered on the 10-K filing date.
[15] The results are essentially the same when standard errors are clustered for filing quarter or filing year.

by about 10% per year. This comparison is consistent with the ever-increasing emphasis on risk disclosures from companies, regulators, and investors. The distribution of Δ*Risk Disclosure* is right-skewed, consistent with firms making large increases to their risk disclosures in certain years. As discussed in Section 2, we differentiate between risk disclosure changes that do and do not coincide with those of companies in the same industry and year. Average (median) Δ*Industry-level RD* is 8.9 (7.0) whereas Δ*Firm-level RD* is 4.8 (0.0).

Figure 1 depicts a monotonic increase in risk disclosures over the sample period. There is a slight increase in 1997 and 1998 coinciding with the passage of FRR No. 48 in 1997. Interestingly, firms did not increase their risk disclosures in 1999 and 2000, before investors experienced significant losses. The risk sentences increased sharply, beginning with the market crash in 2001 and passage of the Sarbanes-Oxley Act of 2002. We also observe a large increase in 2005, coinciding with the SEC's mandate that companies discuss their risk factors concerning operations and future cash flows in the first pages of 10-K filings. Overall, the median number of risk sentences increases nine-fold from an average of 19 sentences in 1994 to 170 sentences in 2007 as compared to a three-fold increase in non-risk sentences from an average of 293 in 1994 and 801 in 2007 (untabulated). In sum, the increases in risk disclosures coincide with related regulatory changes over years, suggesting that our textual analysis code is well-specified.

Panel B of Table 1 presents correlations among key variables. Δ*Risk Disclosure* positively correlates with both firm-level and industry-level changes in risk disclosures. Δ*Risk Disclosure* also positively correlates with Δ*Non-Risk Disclosure*. The univariate correlations of Δ*Risk Disclosure* with dependent variables, Δσ(*Return*), *Log*(*Filing Volume*), Δ*Log*(*Volume*), and σ(*Forecast Revision*), are positive and significant at the 5% level. The correlations suggest that changes in risk disclosures prompt increases in users' risk perceptions. However, it is

difficult to interpret the economic significance of the above associations for two reasons. First, Δ*Risk Disclosure* is a quantitative measure of a qualitative economic construct, making it difficult to establish a threshold of economic significance. Second, we test for the dominant effect of risk disclosures where risk disclosures can be consistent with both the divergent and convergent effects so that the dominant effect will be attenuated. These concerns notwithstanding, we provide multivariate analyses on the above relations, and explore the economic significance by comparing the effect of a one standard deviation change in Δ*Risk Disclosure* to the effect of a one standard deviation change in the market-level control variable for each model.

4.2 Volatility of daily stock returns

Table 2 presents the results of Eq. (1) testing the effect of changes in risk disclosures on changes in daily stock return volatility (Models 1 and 2) and relative change in volatility of negative returns to positive returns (Models 3 and 4). For Model 1, the coefficient on Δ*Risk Disclosure* (multiplied by 1,000 for ease of interpretation) is 0.864 and is significant at the 1% level. In contrast, the change in return volatility is negatively related with the changes in non-risk sentences in the 10-K filing. The difference in the estimated coefficients (0.864 versus -0.101) is statistically significant at the 1% level (untabulated). The stark contrast between the two coefficients is consistent with risk disclosures (non-risk disclosures) in 10-K filings resulting in divergent (convergent) interpretations about future performance. Model 1 also includes a control variable, change in market return volatility, which positively correlates with the change in return volatility.

Model 2 includes additional control variables. The coefficient on Δ*Risk Disclosure*, 0.833, continues to be positive and significant. With respect to the additional control variables,

the change in return volatility is positively associated with the change in the Fog index and the number of managerial forecasts; and is negatively associated with the change in the occurrence of a net loss, company filing return, and absolute value of company filing return.

While the coefficient on Δ*Risk Disclosure* is interpreted as the effect of an additional risk sentence on our dependent variable holding everything else constant, we examine economic significance relative to the effect of change in market return volatility. The change in market return volatility provides a benchmark of a similar scale and measured over the same period as our dependent variable. Specifically, we calculate the effect of a one standard deviation change in Δ*Risk Disclosure* (28.7 sentences) relative to the effect of one standard deviation change in Δ*Market Return Volatility* (0.267) on the dependent variable. When Δ*Risk Disclosure* increases by one standard deviation, Δσ(*Return*) increases by 0.024 (=0.000833*28.7). When Δ*Market Return Volatility* increases by one standard deviation, Δσ(*Return*) increases by 0.091 (=0.339*0.267). This indicates that changes in risk sentences have an effect on return volatility that is 26% (=0.024/0.091) of the effect of comparable changes in market-level return volatility.

For Model 3, the coefficient on Δ*Risk Disclosure* is 0.432 and significant at the 5% level. That is, changes in risk disclosures increase Δ(σ(*Neg Return*)/σ(*Pos Return*)), suggesting that risk disclosures make negative stock returns more volatile. Similar to Model 1, there is a stark contrast between the estimated coefficients on Δ*Risk Disclosure* and Δ*Non-Risk Disclosure*. The risk disclosures (non-risk disclosures) in 10-K filings appear to result in more divergent (convergent) interpretations about negative contingencies. The difference in these estimated coefficients is statistically significant at the 5% level (untabulated). Model 3 also includes a control variable, the change in market return volatility, which positively correlates with Δ(σ(*Neg Return*)/σ(*Pos Return*)).

Model 4 includes additional control variables. The coefficient on Δ*Risk Disclosure*, 0.334, continues to be positive and significant. With respect to the additional control variables, the change in the ratio of volatility of negative returns to that of positive returns is positively associated with the changes in institutional ownership, number of managerial forecasts, and the occurrence of a net loss; and is negatively associated with changes in sales growth, ROA, number of segments, and market return. The negative association between Δ(σ(*Neg Return*)/σ(*Pos Return*)) and Δ(*Market Return*) is expected, because increases in market-level returns will be strongly associated with increases in firm-level returns causing more volatility of positive returns versus negative returns.

When Δ*Risk Disclosure* increases by one standard deviation (28.7 sentences), Δ(σ(*Neg Return*)/σ(*Pos Return*)) increases by 0.010 (=0.000334*28.7), suggesting that the negative return volatility increases by 1% more relative to positive return volatility. When Δ*Market Return Volatility* increases by one standard deviation, Δ(σ(*Neg Return*)/σ(*Pos Return*)) increases by 0.027 (=0.101*0.267). This indicates that the effect of changes in Δ*Risk Disclosure* is 37% (=0.010/0.027) of the effect of comparable changes in Δ*Market Return Volatility*. Therefore, the association between changes in textual risk disclosures and changes in the volatility of negative stock returns is of an economically significant magnitude.

4.3 Trading volume

Models 1 and 2 of Table 3 test how changes in risk disclosures affect *Log*(*Filing Volume*), which is the logarithm of the average three-day trading volume scaled by outstanding shares over the filing date. We modify Eq. (1) with several different control variables. The base model (Model 1) includes a control variable for the normal level of trading volume over the three month period ending 60 trading days before the filings, *Log*(*Non-Filing Volume*). Consistent

with Garfinkel (2009), we also include *Filing Return* and *Absolute Filing Return* to control for the expected trading volume resulting from stock prices changes. We also control for market-level trading volume by including *Log*(*Market Volume*), the logarithm of CRSP firms' value-weighted average three-day trading volume scaled by outstanding shares over the filing date.

In Model 1, the coefficient on Δ*Risk Disclosure* is 0.489 and significant at the 5% level, indicating that changes in risk disclosures are positively associated with trading volume during the three-day filing date period. *Log*(*Non-Filing Volume*) is positively associated with trading volume as expected, because this variable serves as a benchmark for the normal level of trading volume. *Filing Return*, *Absolute Filing Return*, and *Market Volume* are all significantly positively associated with trading volume as expected. Model 2 includes additional control variables to rule out the possibility that other information is driving our results. The coefficient on Δ*Risk Disclosure* continues to be positive and significant. With respect to the additional control variables, *Log*(*Filing Volume*) is positively associated with the change in institutional ownership, and change in the number of managerial forecasts; and is negatively associated with the change in ROA.

With respect to economic significance, a one standard deviation increase in Δ*Risk Disclosure* increases *Log*(*Filing Volume*) by 0.014 (=0.000485*28.7), holding everything else constant. We compare this to the effect of a one standard deviation increase in *Log*(*Market Volume*) on *Log*(*Filing Volume*), because *Log*(*Market Volume*) is of a similar scale and measured over the same time period as our dependent variable. When *Log*(*Market Volume*) increases by one standard deviation (0.290), *Log*(*Filing Volume*) increases by 0.057 (=0.195*0.290). This indicates that changes in risk sentences have an effect on *Log*(*Filing Volume*) that is 25% (=0.014/0.057) of the effect of comparable changes in market level trading volume.

Models 3 and 4 of Table 3 test how changes in risk disclosures affect Δ*Log*(*Volume*), the change in the logarithm of average daily trading volume from the 60 trading-day period before to the 60 trading-day period after the 10-K filings. In the base model (Model 3), we include Δ*Non-Risk Disclosure, Log*(*Non-Filing Volume*), *Absolute Filing Return, Filing Return,* and Δ*Log*(*Market Volume*) as control variables. Δ*Log*(*Market Volume*) is the change in the logarithm of CRSP firms' value-weighted average trading volume scaled by outstanding shares from the 60 trading-day period before to the 60 trading-day period after the filings. The coefficient on Δ*Risk Disclosure* is 0.406 and significant at the 1% level, indicating that changes in risk disclosure are positively associated with changes in trading volume. *Log*(*Non-Filing Volume*) is negatively associated with changes in trading volume, which is consistent with firms with generally high levels of trading volume experiencing a lower percentage change in their trading volume after 10-K filings. Δ*Log*(*Market Volume*) is significantly positively associated with the change in trading volume as expected. *Filing Return* and *Absolute Filing Return* are also significantly positively associated with the change in trading volume. When we include additional control variables in Model 4, the coefficient on Δ*Risk Disclosure* continues to be positive and significant. With respect to the additional control variables, Δ*Log*(*Volume*) is positively associated with the change in institutional ownership and change in the number of managerial forecasts. Δ*Log*(*Volume*) is negatively associated with the change in ROA and change in the occurrence of a net loss.

With respect to economic significance, a one standard deviation increase in Δ*Risk Disclosure* is associated with change in Δ*Log*(*Volume*) of 0.012 (=0.000411*28.7), holding everything else constant, while a one standard deviation change in Δ*Log*(*Market Volume*) is associated with a change in Δ*Log*(*Volume*) of 0.071 (=0.727*0.097). This indicates that changes

in risk sentences have an effect on Δ*Log*(*Volume*) that is 17% (=0.012/0.071) of the effect of comparable changes in market level trading volume, holding the effects of other control variables constant.

Overall, the trading volume results in Table 3 triangulate our findings on stock return volatility. The changes in risk disclosures are associated with economically significant changes in trading volume. Increases in risk disclosures appear to prompt investors to differentially revise their prior beliefs and, in turn, increase their trading during the filing period and after the filing. At the same time, increases in risk disclosures appear to reduce confidence of individual investors in their predictions so that they are more likely to trade with the arrival of new information.

4.4 Volatility of analyst forecast revisions

The above evidence of increased trading volume around the filings can also result from risk disclosures converging investors' beliefs that were divergent prior to the filing (Bamber and Cheon 1995). To more clearly explain whether users' interpretations converge or diverge, we examine how sell-side equity analysts, who are typically superior users of 10-K information, react to risk disclosures. We examine the relation between changes in risk disclosure and the standard deviation of individual analysts' forecast revisions around the filings, σ(*Forecast Revision*), after controlling for changes in non-risk disclosures and market return volatility. Table 4, Model 1 presents the regression results. The coefficient on Δ*Risk Disclosure* is 0.008 and is significant at the 10% level, suggesting that risk disclosures prompt analysts to make more volatile changes to their one-year-ahead earnings forecasts.[16] The other control variables are not

[16] In order to fully capture analysts' differential interpretations around 10-K filings, we try two dependent variable alternatives to σ(*Forecast Revision*). First, Δ*Forecast Dispersion* is the difference in standard deviation of one-year-ahead earnings forecasts issued during the first two months after the filings and during the last two months before the filings. Second, *KP Forecast Divergence* is the proportion of all pairs of analysts' forecast revisions that diverge

significant at conventional levels. Model 2 includes additional control variables of Eq. (1). We also include *Number of Analysts* as a control variable, because the number of analysts can affect the level of noise in the standard deviation of forecast revisions. The coefficient on Δ*Risk Disclosure* is 0.009 and is significant at the 5% level. With respect to the control variables, volatility of forecast revisions is positively associated with the change in ROA, change in the occurrence of a net loss, and absolute value of the filing return; and is negatively associated with changes in non-risk sentences, sales growth, and number of business segments. The stark contrast between the coefficients on changes in risk disclosures and non-risk disclosures in the 10-K filings is similar to the results in Table 2, consistent with risk disclosures (non-risk disclosures) in 10-K filings resulting in divergent (convergent) interpretations about future performance.

With respect to economic significance, a one standard deviation increase in Δ*Risk Disclosure* is associated with a change in σ(*Forecast Revision*) of 0.00026 (=0.000009*28.7), holding everything else constant. There is no clear control variable in this model to compare with the effect of Δ*Risk Disclosure*, so we consider this effect relative to the average level of volatility in forecast revisions. The average σ(*Forecast Revision*) is 0.005. The effect of a one standard deviation increase in Δ*Risk Disclosure* is 5.2% (=0.00026/0.005) of the average σ(*Forecast Revision*). Overall, analysts appear to issue more dispersed forecast revisions when changes in risk disclosures are higher.

4.5 Firm-level versus industry-level risk disclosures

We reperform the above tests by dividing Δ*Risk Disclosure* into Δ*Firm-Level RD* and Δ*Industry-Level RD*. Table 5, Panel A provides results for regressions where the dependent

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from each other (Kandel and Pearson 1995). While these two variables correlate positively with Δ*Risk Disclosure*, the related coefficient estimates in Eq. (1) are not significant.

variables are the change in return volatility, Δσ(*Return*), the relative change in negative return volatility, Δ(σ(*Neg Ret*)/σ(*Pos Ret*)), filing trading volume, *Log*(*Filing Volume*), change in trading volume, ΔLog(*Volume*), and volatility of forecast revisions, σ(*Forecast Revision*). The full set of control variables are included in the regressions but not in the table for brevity. The coefficients on Δ*Industry-Level RD* are positive and significant in all models. The coefficients on Δ*Firm-Level RD* are positive and significant when the dependent variables are the change in return volatility, relative change in negative return volatility, and filing trading volume. Most importantly, in all models, the estimated coefficients on Δ*Industry-Level RD* are significantly higher than those on Δ*Firm-Level RD*. The results collectively suggest that industry-level changes in risk disclosures are significantly more effective than firm-level changes in affecting investors' risk perceptions. This is consistent with the criticism that risk disclosures lack useful firm-specific details.

4.6 Risk disclosures that emphasize negative outcomes

In an exploratory analysis, we modified our analyses based on whether risk-related sentences have additional negative tone. Risk disclosures are expected to provide information about downside risk in general. The literature on prospect theory predicts and documents that individuals react to prospects of losses asymmetrically (Kahneman and Tversky 1979; Koonce et al. 2005). Therefore, it is likely that risk disclosures that emphasize negative outcomes diverge investor beliefs more. However, managers can emphasize negative–but not more useful–risk statements about broad risk factors in order to avoid legal consequences for failing to reveal negative news. Therefore, we examine whether a negative emphasis in risk disclosures impacts users' risk perceptions differently than other risk disclosures. We divide Δ*Risk Disclosure* into Δ*Negative RD* and Δ*Other RD*. Δ*Negative RD* is equal to the change in the number of risk

sentences with a negative tone, i.e., those that involve any of the following keywords: negative*, material*, adverse*, damage*, destroy*, loss, harm, catastroph*, tragic, destruct*, serious, and hamper. Δ*Other RD* is equal to change in the number of risk sentences that do not involve the above keywords. We do not find consistent evidence that risk disclosures with a negative tone affect our dependent variables differently than other risk disclosures. This null result is possibly due to measurement error in our classification of negative tone. We supplement this test by subjectively reviewing a random sample of 20 risk disclosures with a negative tone and we find there to be a similar number of those that appear boilerplate and informative. An area of potential future research is to refine our measure of textual risk disclosure to capture more specific information with regard to the tone or severity of these disclosures.

4.7 Alternative explanations

Annual reports may not reveal contingencies and risk factors but only correlate with changes in company risks that users learn from other information sources. This alternative explanation is not likely driving our results for three reasons. First, we investigate changes in investor and analyst activity during short-window periods immediately before and after 10-K filing dates, so other information sources would have to be concentrated around 10-K filing dates to explain the documented changes in analyst and investor activity. Second, our research design controls for changes in other information sources such as institutional investors, management earnings forecasts, sales growth, ROA, number of business segments, and reporting of losses. Non-risk information in the 10-K filings as well as the readability of the annual report is also controlled for through the total number of non-risk sentences and the Fog index. We also control for changes in economic risk factors using such variables as the change in the market return,

volatility of the market return, and market level trading volume.[17] Third, we find consistent results in the short-window trading volume test where the release of other risk-related information is a less plausible explanation. Nevertheless, this alternative explanation, even if valid, suggests that company risk disclosures, though not timely, are not boilerplate and do reflect corporate risk exposures. While this interpretation affects some of our inferences, we argue it does not lessen the contribution of our study in understanding the information content of textual risk disclosures in 10-K filings.

In untabulated tests, we test whether increases and decreases in risk disclosures affect users' risk perceptions differently. We include in the tests from Tables 2 to 4, an interaction term of Δ*Risk Disclosure* with an indicator variable for negative Δ*Risk Disclosure*. We also include in the tests of Table 5, which use both firm-level and industry-level risk disclosures, an interaction term of Δ*Firm-Level RD* with an indicator variable for negative Δ*Firm-Level RD*. We do not find any consistent evidence that annual increases and decreases in risk disclosures have differential effects on investors' risk perceptions. This finding mitigates concerns about non-linearity in the relation between risk disclosures and risk perceptions.

We also use Li's (2006) definition of risk sentiment (i.e., keywords of "risk" and "uncertainty") to ensure our results are not limited to our specific keyword list. The results are generally consistent. Finally, we fail to find a time trend in the relative informativeness of risk disclosures during our sample period suggesting that increases in risk disclosures over time in response to mandates and litigation concerns do not appear to result in less informative risk disclosures.

[17] In untabulated tests, we include firm fixed effects and find similar results as those that are reported.

5 Conclusion

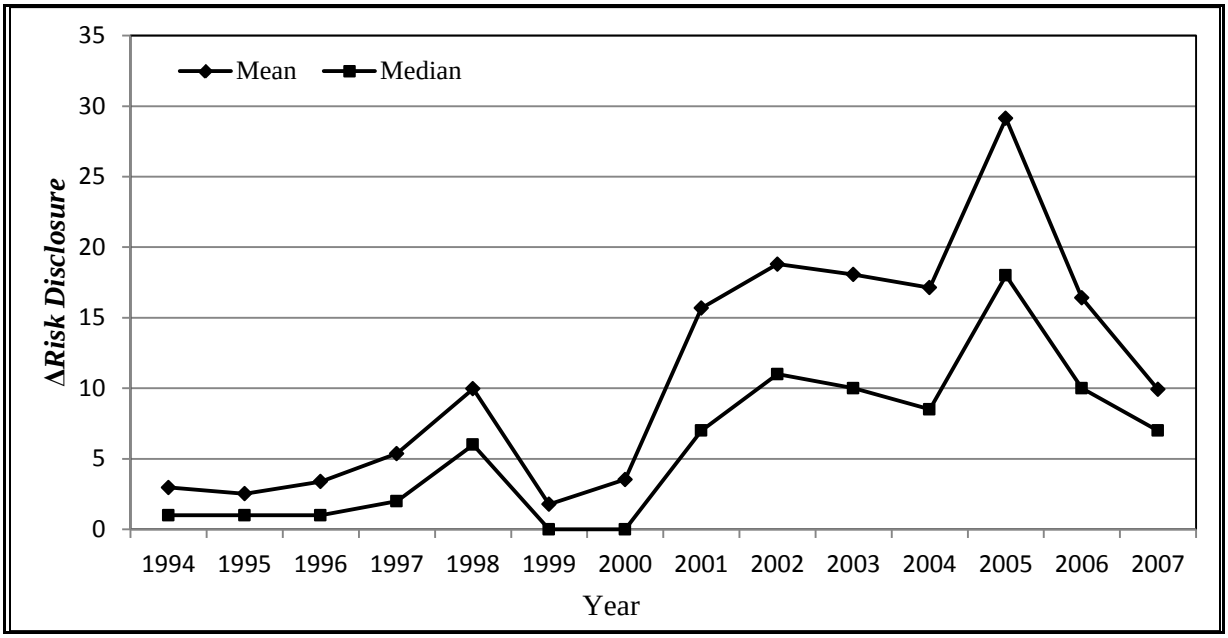
Textual risk disclosures in 10-K filings have increased to a greater extent than non-risk textual disclosures during our sample period; this trend is likely to continue given that regulators tend to tighten risk disclosure standards in response to economic crises. It is thus important to assess how investors benefit from risk disclosures. In this paper, we test how annual changes in textual risk disclosures in 10-K filings impact users' risk perceptions, as measured by investor and analyst activities within the immediate period before and after the 10-K filings. The empirical tests use a changes specification, and thus are relatively void of empirical issues such as correlated omitted factors and reverse causality.

We find that annual changes in risk disclosures are significantly and positively associated with changes in daily stock return volatility, changes in relative volatility of negative daily returns, filing volume, changes in trading volume, and changes in volatility of forecast revisions. The results are economically significant when compared to the effect of other market-based variables. These results reject the null argument that risk disclosures are boilerplate. Furthermore, consistent with the divergence argument dominating the convergence argument, company risk disclosures appear to introduce unknown contingencies and risk factors rather than only update information about known risks. Our findings contrast with those in prior literature, which generally documents a negative correlation between company disclosures and divergence in market participants' beliefs.

We close by suggesting avenues for future research. In Li's (2010a) survey of textual analysis literature, he points to a vacuum of research that ties large-sample textual disclosures to incentives of managers due to compensation, contracting, and capital market considerations. We agree with this assessment, particularly in the context of risk disclosures. Managerial incentives

are likely to have a stronger impact on how managers communicate information on “what may happen”, which is harder to assess by managers and harder to monitor by outsiders than information about “what has happened” or “what will happen”. Future research can examine the effect of various managerial incentives on risk disclosures and informativeness of risk disclosures. Furthermore, the impact of risk disclosures are hardly limited to equity markets, given that companies are required to provide information about credit, interest, and currency risks. Future research can investigate how debt markets or credit rating agencies respond to risk disclosures. Lastly, we show that an element of risk disclosure that is particularly important to users, idiosyncratic firm-specific disclosures, consistently draw less response from investors. That idiosyncratic risk disclosures are less informative is worthy of further research and important for standard setters.

Figure 1 Change in the number of risk sentences by year



The figure presents the mean and median change in the number of risk sentences by year. The sample includes 28,110 observations collected from 10-K filings between the years 1994 and 2007.

Appendix A Descriptive information of risk disclosure measure

Table A1 Average number of risk-related keywords in an annual report

	Average number of risk-related keywords	Average change in number of risk-related keywords
may	41	5
could	21	3
can/cannot	18	2
risk*	18	2
subject to	15	2
affect	14	2
potential*	8	1
depend*	8	1
expos*	6	1
hedg*	4	1
fluctuat*	4	1
uncertain*	4	1
possibl*	3	0
vary*/varies	3	0
might	1	0
likely to	1	0
influenc*	1	0
susceptible	0	0
Total	170	22

* denotes that suffixes are allowed.

Table A2 Examples of new risk sentences in 10-K filings

Keyword	Sentence
May	As a result of the significant investment required with a product launch, we believe that our share of the co-promotion split from the sale of Tarceva (TM) in the United States **may** not be profitable in the near term (OSI Pharmaceuticals Inc., 12/04/2004). If the Company cannot comply with the requirements in its warehouse credit facilities, then the lenders **may** require it to immediately repay all of the outstanding debt under its facilities (AmeriCredit Corporation, 09/25/2001).
Could	The Factiva business **could** be negatively affected by significant investments that its commercial competitors might make in their aggregation products, by dynamics in the publishing market that rendered publishers of newspapers, business journals and other periodicals less willing to license their content for inclusion in the Factiva services (or to demand higher royalties for such licenses), or by an improvement in the quality of the business news and information that is available for free on the internet or in its presentation or accessibility (Dow Jones Inc., 3/1/2007). Such royalties approximate $44 million for 1997…A termination of the Ramada License Agreements would result in the loss of the income stream from franchising the Ramada brand names and **could** result in the payment by the Company of liquidated damages equal to three years of license fees (Cendant Corporation, 03/31/1998).

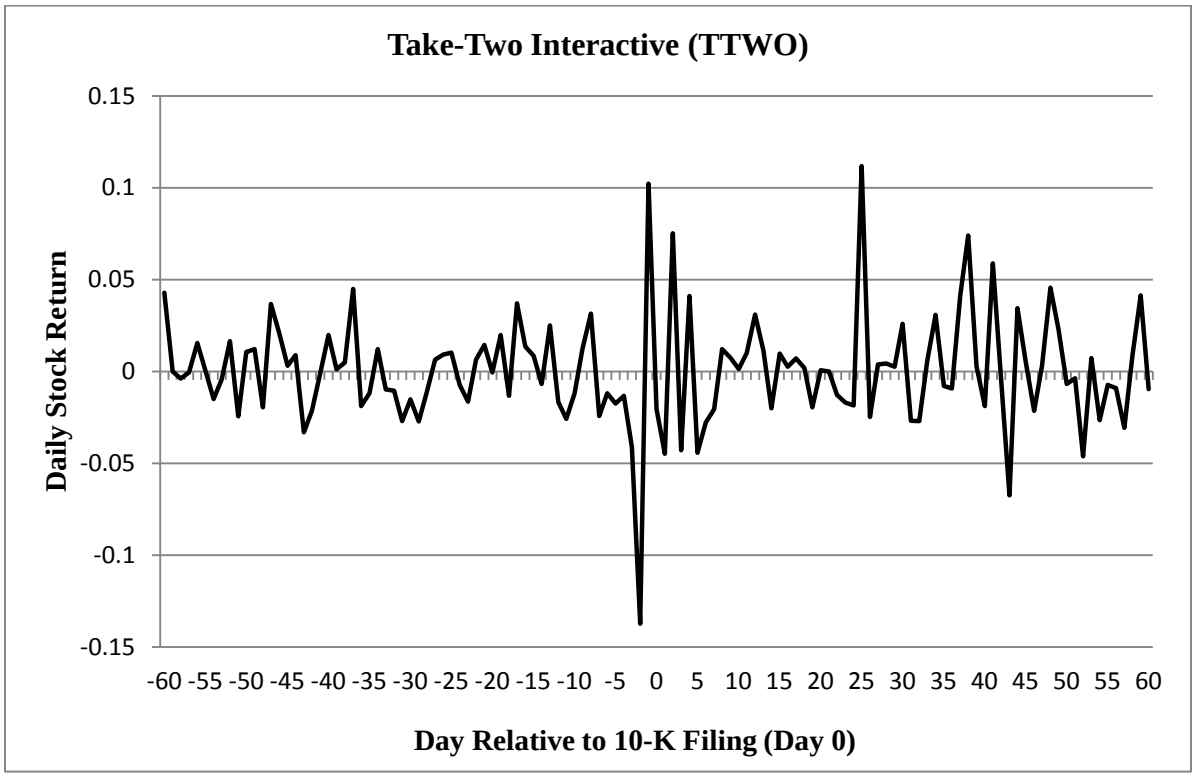
Can/ Cannot	We strongly disagree with the position taken by those insurers and continue to believe that the EchoStar IV insurance claim will be resolved in a manner satisfactory to us. However, we **cannot** assure you that we will receive the amount claimed or, if we do, that we will retain title to EchoStar IV with its reduced capacity (Echostar Communications Corp, 03/13/2000). There **can** be no assurance that the volume of searches conducted, the amounts bid by advertisers for search listings or the number of advertisers that bid on the Overture service will not vary widely from period to period (Yahoo Inc., 02/27/2004).
Risk	The Company bears the **risk** of cost overruns and inflation in connection with fixed-price engagements, and as a result, any of these engagements may be unprofitable (Aspen Technologies, 09/28/1998). The foregoing discussion of our IPR&D projects, and in particular the following table and subsequent paragraphs regarding the future of these projects, our additional product programs and our process technology program include forward-looking statements that involve **risks** and uncertainties, and actual results may vary materially (Genentech Inc., 03/04/2002).
Subject to	Completion of the merger is **subject to** the satisfaction of various conditions, including adoption of the merger agreement by holders of a majority of the outstanding shares of our common stock, expiration or termination of applicable waiting periods under the HSR Act (the FTC granted early termination of the applicable waiting period on May 18, 2007) and other non-U.S. competition laws, and other customary closing conditions described in the merger agreement. (Harman International Industries, 08/29/2007). Part of this portfolio includes minority equity investments in several publicly traded companies, the values of which are **subject to** market price volatility. For example, as a result of recent market price volatility of our publicly traded equity investments, we experienced a $111 million after-tax unrealized loss during the third quarter of fiscal 2000 and a $1.83 billion after-tax unrealized gain during the fourth quarter of fiscal 2000 on these investments (Cisco, 09/29/2000).
Affect	This increase has had an adverse impact on our results of operations and will continue to adversely **affect** our results of operations unless our customers share in these increased costs (Visteon, 03/16/2005). Our strategy over the past several years with respect to real estate has been to reduce our holdings of excess real estate. In line with this strategy, we anticipate the exit of facilities, which may **affect** net income (NCR Corporation, 03/07/2005).
Uncertain	The transition to retail competition continues to be highly **uncertain**, driven by the development of a relatively young wholesale market and the different approaches to retail competition taken by state regulators and legislators (NV Energy, 3/20/2002). NYRA's recent filing for reorganization under Chapter 11 has introduced additional **uncertainties**, but we remain committed to the development once these uncertainties are resolved (MGM, 02/28/2007).

Table A3 Average number of total sentences and risk sentences in the sections of 10-K filings

Section	Title	Total Sentences	Δ Total Sentences	Total Risk Sentences	Δ Total Risk Sentences
Part 1					
1	Business	214	13	43	2
1A	Risk Factors	30	15	18	10
1B	Unresolved Staff Comments	1	0	0	0
2	Properties	15	-1	1	0
3	Legal Proceedings	14	0	2	0
4	Submission of Matters to a Vote of Security Holders	19	-1	1	0
Part 2					
5	Market for Company's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities	13	0	2	0
6	Selected Financial Data	11	0	1	0
7	Management's Discussion and Analysis of Financial Condition and Results of Operation	213	17	37	2
7A	Quantitative and Qualitative Disclosures about Market Risk	11	1	4	0
8	Financial Statements and Supplementary Data	151	16	19	2
9	Changes in and Disagreements with Accountants on Accounting and Financial Disclosure	20	2	3	0
Part 3					
10	Directors, Executive Officers and Corporate Governance	20	-1	0	0
11	Executive Compensation	10	0	1	0
12	Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters	6	0	1	0
13	Certain Relationships and Related Transactions, and Director Independence	4	0	0	0
Total		756	57	130	15

The change in total sentences and risk sentences in this table are slightly different than those used in the empirical analyses, because this table does not use approximately 20% of the sample due to these 10-K filings lacking all or some of the section title numbers.

Appendix B Anecdote



Appendix B presents the daily stock returns for the 60-day period before and after the January 1, 2006 10-K filing of Take-Two Interactive. The standard deviation of returns ($\Delta\sigma(\textit{Return})$) increased from 2.6% to 3.1%. The example is based on a news media article citing risk disclosures in the company's 10-K filing (Consumer Electronics Daily, February 2, 2006).

Article Excerpts

"Take-Two Interactive 'reached an agreement in principle' to retain, for 3 years, key employees at its Rockstar game studio responsible for the hit Grand Theft Auto series, the company disclosed in its 10-K filing at the SEC. But Take-Two said the 'compensation arrangements could result in increased expenses and have a negative impact on our operating results.'"

"Take-Two warned in the 10-K…that a failure to reach a definitive deal with the Rockstar employees and if one or more of them leave Take-Two, 'we may lose additional personnel, experience material interruptions in product development and delays in bringing products to market'. It said that 'could have a material adverse effect on our operating results.'"

"Take-Two also warned investors that its publishing and distribution activities require significant cash resources [and it] may be required to seek debt or equity financing to fund the cost of continued expansion."

Appendix C Variable definitions

10-K filing variables	
$\Delta Risk\ Disclosure_{i,t}$	The change in the number of sentences that contain risk keywords between firm *i*'s 10-K filings for fiscal years *t* and *t*-1.
$\Delta Industry\text{-}Level\ RD_{i,t}$	The industry and year-median of $\Delta Risk\ Disclosure_{i,t}$, where industry is defined by 4-digit SIC codes.
$\Delta Firm\text{-}Level\ RD_{i,t}$	The industry and year median-adjusted value of $\Delta Risk\ Disclosure_{i,t}$, calculated as $\Delta Risk\ Disclosure_{i,t}$ - $\Delta Industry\text{-}Level\ RD_{i,t}$.
$\Delta Non\text{-}Risk\ Disclosure_{i,t}$	The change in the number of sentences that do not contain risk keywords between firm *i*'s 10-K filings for fiscal years *t* and *t*-1.
Dependent variables	
$\Delta\sigma(Return)_{i,t}$	The change in the standard deviation of firm *i*'s daily stock returns between the 60 trading-day period before and the 60 trading-day period after firm *i*'s 10-K filing for fiscal year *t*, multiplied by 100. The calculation excludes the three-day period [-1, 1] surrounding the 10-K filing.
$\Delta(\sigma(Neg\ Return)/\sigma(Pos\ Return))_{i,t}$	The change in the ratio of $\sigma(Neg\ Return)_{i,t}/\sigma(Pos\ Return)_{i,t}$, between the 60 trading-day period before and the 60 trading-day period after firm *i*'s 10-K filing for fiscal year *t*. $\sigma(Neg\ Return)_{i,t}$ ($\sigma(Pos\ Return)_{i,t}$) is the standard deviation of daily stock returns during trading days with negative (positive) returns where days with positive (negative) returns are valued at zero. The calculation excludes the three-day period [-1, 1] surrounding the 10-K filing.
$Log(Filing\ Volume)_{i,t}$	The natural logarithm of firm *i*'s average daily trading volume divided by outstanding shares in the three-day window surrounding firm *i*'s 10-K filing for fiscal year *t*.
$\Delta Log(Volume)_{i,t}$	The change in firm *i*'s natural logarithm of the average daily trading volume divided by outstanding shares between the 60 trading-day period before and the 60 trading-day period after firm *i*'s 10-K filing for fiscal year *t*. The calculation excludes the three-day period [-1, 1] surrounding the 10-K filing.
$\sigma(Forecast\ Revision)_{i,t}$	The standard deviation of analyst forecast revisions of firm *i*'s fiscal year *t*+1earnings. The forecast revisions are calculated as individual analysts' first forecasts during the first two months after the 10-K filing for fiscal year *t* minus their last forecasts during the last two months before the filing. The calculation excludes forecasts made during the three-day period [-1, 1] surrounding the 10-K filing.
Control variables	
$\Delta Market\ Return\ Volatility_{i,t}$	The change in the standard deviation of the value-weighted market return between the 60 trading-day period before and the 60 trading-day period after firm *i*'s 10-K filing for fiscal year *t*, multiplied by 100. The calculation

	excludes the three-day period [-1, 1] surrounding the 10-K filing.
$Log(Market\ Volume)_{i,t}$	The logarithm of CRSP firms' value-weighted three-day trading volume scaled by outstanding shares surrounding firm *i*'s 10-K filing for fiscal year *t*. The measure is weighted by firms' market capitalization at the beginning of the fiscal year.
$\Delta Log(Market\ Volume)_{i,t}$	The change in the logarithm of CRSP firms' value-weighted trading volume scaled by outstanding shares from the 60 trading-day period before to the 60 trading-day period after firm *i*'s 10-K filing. The measure is weighted by firms' market capitalization at the beginning of the fiscal year. The calculation excludes the three-day period [-1, 1] surrounding the 10-K filing.
$\Delta Fog\ Index_{i,t}$	The change in the Fog Index of firm *i*'s 10-K filings between fiscal years *t* and *t*-1. The Fog Index is calculated as (average words per sentence + percent of complex words) * 0.4 (Li 2008).
$\Delta Institutional\ Ownership_{i,t}$	The change in the percentage of institutional ownership between the end of the last fiscal quarter before and the end of the first fiscal quarter after firm *i*'s 10-K filing for fiscal year *t*.
$\Delta Managerial\ Forecast_{i,t}$	The change in the number of management forecasts between the last two months before and the first two months after firm *i*'s 10-K filing for fiscal year *t*.
$\Delta Sales\ Growth_{i,t}$	The change in the seasonally-adjusted sales growth between the fourth quarter of fiscal year t and the first quarter of fiscal year *t*+1, where seasonally-adjusted quarterly sales growth is calculated as sales in quarter *q* divided by sales in quarter *q*-4.
$\Delta ROA_{i,t}$	The change in the seasonally-adjusted net income between the fourth quarter of fiscal year *t* and the first quarter of fiscal year *t*+1, where seasonally-adjusted net income is calculated as the change in net income before extraordinary items from the same prior fiscal quarter divided by total assets.
$\Delta Segments_{i,t}$	The change in the number of firm *i*'s business segments between fiscal years *t*-1 and *t*.
$\Delta Loss_{i,t}$	An indicator variable equal to one if firm *i* does not report a net loss in the fourth fiscal quarter of fiscal year t and reports a net loss in the first fiscal quarter of fiscal year *t*+1.
$Filing\ Return_{i,t}$	Firm *i*'s stock returns over the three-day period [-1, 1] surrounding firm *i*'s 10-K filing for fiscal year *t*.
$Absolute\ Filing\ Return_{i,t}$	The absolute value of firm *i*'s return over the three-day period [-1, 1] surrounding firm *i*'s 10-K filing for fiscal year *t*.
$Log(Non\text{-}Filing\ Volume)_{i,t}$	The average natural logarithm of firm *i*'s daily trading volume divided by outstanding shares over the three month period ending 60 trading days prior

	to firm *i*'s 10-K filing for fiscal year *t*.
$\Delta Market\ Return_{i,t}$	The change in the value-weighted market returns between the 60 trading-day period before and the 60 trading-day period after firm *i*'s 10-K filing for fiscal year *t*.
$Number\ of\ Analysts_{i,t}$	The number of analysts that issue forecasts of fiscal year *t*+1 earnings during the two months before and after firm *i*'s 10-K filing for fiscal year *t*.

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Table 1 Sample

Panel A: Descriptive statistics

	N	Mean	Median	Standard Deviation	Lower Quartile	Upper Quartile
10-K filing variables						
Δ*Risk Disclosure*	28,110	13.749	6.000	28.744	-1.000	21.000
Δ*Industry-Level RD*	28,110	8.863	7.000	10.485	1.500	13.000
Δ*Firm-Level RD*	28,110	4.799	0.000	26.302	-6.500	10.000
Δ*Non-Risk Disclosure*	28,110	45.767	23.000	141.298	-9.000	82.000
Dependent variables						
Δσ(*Return*)	28,110	-0.010	-0.006	1.185	-0.490	0.451
Δ(σ(*Neg Return*)/ σ(*Pos Return*))	28,110	0.005	0.003	0.437	-0.244	0.253
Log(*Filing Volume*)	28,110	-1.448	-1.179	1.650	-2.167	-0.381
Δ*Log*(*Volume*)	28,110	0.045	0.018	0.501	-0.223	0.280
σ(*Forecast Revision*)	4,502	0.005	0.002	0.013	0.001	0.005
Control variables						
Δ*Market Return Volatility*	28,110	0.021	0.045	0.267	-0.109	0.174
Log(*Market Volume*)	28,110	-0.411	-0.372	0.290	-0.587	-0.243
Δ*Log*(*Market Volume*)	28,110	0.007	0.030	0.097	-0.057	0.066
Δ*Fog Index*	28,110	-0.110	0.000	3.695	-1.359	1.182
Δ*Institutional Ownership*	28,110	0.007	0.001	0.053	-0.011	0.025
Δ*Managerial Forecast*	28,110	-0.070	0.000	0.818	0.000	0.000
Δ*Sales Growth*	28,110	-0.008	-0.003	0.310	-0.084	0.075
Δ*ROA*	28,110	-0.002	0.000	0.061	-0.008	0.007
Δ*Segments*	28,110	0.044	0.000	0.370	0.000	0.000
Δ*Loss*	28,110	0.065	0.000	0.247	0.000	0.000
Filing Return	28,110	-0.001	-0.001	0.053	-0.023	0.020
Absolute Filing Return	28,110	0.036	0.021	0.043	0.009	0.045
Log(Non-Filing Volume)	28,110	-1.513	-1.246	1.574	-2.263	-0.461
Δ*Market Return*	28,110	0.010	-0.006	0.104	-0.050	0.063
Number of Analysts	4,502	8.903	7.000	4.237	6.000	11.000

Panel B: Pearson (top) and Spearman (bottom) correlation coefficients

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1)	Δ*Risk Disclosure*		**0.92**	**0.40**	**0.62**	**0.01**	0.00	**0.10**	**0.02**	**0.05**
(2)	Δ*Firm-Level RD*	**0.83**		**0.03**	**0.60**	0.01	0.00	**0.07**	0.00	**0.04**
(3)	Δ*Industry-Level RD*	**0.44**	**-0.03**		**0.19**	**0.03**	**0.02**	**0.10**	**0.07**	**0.03**
(4)	Δ*Non-Risk Disclosure*	**0.58**	**0.49**	**0.25**		0.01	**-0.01**	**0.06**	**0.01**	**0.03**
(5)	Δσ*(Return)*	**0.02**	0.01	**0.04**	0.01		**-0.09**	**-0.06**	**0.30**	**-0.08**
(6)	Δ(σ(*Neg Return*)/ σ(*Pos Return*))	0.00	-0.01	0.01	-0.01	**-0.10**		**0.02**	**-0.05**	-0.03
(7)	*Log*(*Filing Volume*)	**0.12**	**0.07**	**0.15**	**0.08**	**-0.05**	**0.04**		**-0.06**	**0.11**
(8)	Δ*Log*(*Volume*)	**0.04**	0.00	**0.08**	**0.02**	**0.28**	**-0.05**	**-0.04**		0.00
(9)	σ(*Forecast Revision*)	**0.08**	**0.05**	**0.12**	**0.04**	**-0.03**	**-0.03**	**0.19**	0.01	

Panel A presents descriptive statistics of the variables used in the tests. Panel B presents the correlation coefficients for the main variables; bold numbers indicate statistical significance at the 5% level. Variable descriptions appear in Appendix C. All continuous variables are winsorized at the 1st and 99th percentile except for indicator variables.

Table 2 Change in volatility of stock returns

	(1)		(2)		(3)		(4)	
	Δσ(*Return*)		Δσ(*Return*)		Δ(σ(*Neg Return*)/ σ(*Pos Return*))		Δ(σ(*Neg Return*)/ σ(*Pos Return*))	
	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic
Intercept	-0.032	-1.17	-0.012	-0.52	-0.002	-0.11	0.002	0.17
Δ*Risk Disclosure (x1,000)*	0.864	2.91 ***	0.833	3.03 ***	0.432	2.37 **	0.334	2.35 **
Δ*Non-Risk Disclosure (x1,000)*	-0.101	-1.95 **	-0.087	-1.66 *	-0.072	-2.23 **	-0.056	-1.97 **
Δ*Market Return Volatility*	0.355	3.42 ***	0.339	2.77 ***	0.152	4.51 ***	0.101	5.08 ***
Δ*Fog Index (x1,000)*			2.880	1.75 *			0.064	0.13
Δ*Institutional Ownership*			-0.018	-0.14			0.484	5.27 ***
Δ*Managerial Forecast*			0.057	8.09 ***			0.014	4.40 ***
Δ*Sales Growth*			-0.019	-0.90			-0.019	-2.42 **
Δ*ROA*			-0.007	-0.05			-0.217	-4.14 ***
Δ*Segments*			0.009	0.40			-0.031	-3.12 ***
Δ*Loss*			-0.046	-1.93 *			0.058	7.48 ***
Filing Return			-0.664	-4.29 ***			0.040	0.58
Absolute Filing Return			-0.340	-2.22 **			-0.037	-0.41
Δ*Market Return*			-0.156	-0.39			-0.573	-7.17 ***
N	28,110		28,110		28,110		28,110	
Adj. R^2	1.4%		2.0%		1.4%		5.3%	

This table tests the association between changes in 10-K risk disclosures and changes in the volatility of stock returns and negative stock returns between the 60 trading-day period after and the 60 trading-day period before the 10-K filing dates. Variable descriptions appear in Appendix C. All continuous variables are winsorized at the 1st and 99th percentile, and influential observations with studentized t-statistics greater than two are excluded. Standard errors are heteroskedasticity-adjusted and are clustered for firm and filing month. *, **, and *** denote 10%, 5%, and 1% significance levels, respectively.

Table 3 Trading volume

	(1)		(2)		(3)		(4)	
	Log(*Filing Volume*)		*Log*(*Filing Volume*)		Δ*Log*(*Volume*)		Δ*Log*(*Volume*)	
	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic	Coefficient	*t*-statistic
Intercept	-0.150	-5.27 ***	-0.151	-5.19 ***	-0.075	-7.67 ***	-0.074	-7.56 ***
Δ*Risk Disclosure (x1,000)*	0.489	2.24 **	0.485	2.29 **	0.406	2.91 ***	0.411	2.94 ***
Δ*Non-Risk Disclosure (x1,000)*	-0.016	-0.55	-0.012	-0.41	0.005	0.24	0.009	0.40
Log(*Non-Filing Volume*)	0.886	193.38 ***	0.886	190.87 ***	-0.054	-12.76 ***	-0.054	-12.65 ***
Filing Return	0.719	6.21 ***	0.716	6.13 ***	0.241	3.11 ***	0.236	2.98 ***
Absolute Filing Return	4.192	16.74 ***	4.203	16.86 ***	0.484	2.35 **	0.495	2.37 **
Log(*Market Volume*)	0.196	4.15 ***	0.195	4.13 ***				
Δ *Log*(*Market Volume*)					0.737	5.57 ***	0.727	5.47 ***
Δ*Fog Index (x1,000)*			0.292	0.30			0.711	1.28
Δ*Institutional Ownership*			0.167	1.88 *			0.210	3.54 ***
Δ*Managerial Forecast*			0.011	3.78 ***			0.015	6.89 ***
Δ*Sales Growth*			-0.006	-0.43			0.013	1.62
Δ*ROA*			-0.118	-1.65 *			-0.145	-3.59 ***
Δ*Segments*			-0.007	-0.69			0.000	0.03
Δ*Loss*			-0.016	-1.04			-0.028	-2.94 ***
N	28,110		28,110		28,110		28,110	
Adj. R^2	85.2%		85.3%		7.0%		7.6%	

The first and second models test the association between changes in 10-K risk disclosures and the trading volume over the three-day window surrounding the 10-K filing [-1, 1]. The third and fourth models test the association between changes in 10-K risk disclosures and changes in trading volume surrounding 10-K filing dates from the 60 trading-day period before the filing to the 60 trading-day period after the filing. Variable definitions appear in Appendix C. All continuous variables are winsorized at the 1st and 99th percentile, and influential observations with studentized t-statistics greater than two are excluded. Standard errors are heteroskedasticity-adjusted and are clustered for firm and filing month. *, **, and *** denote 10%, 5%, and 1% significance levels, respectively.

Table 4 Volatility of forecast revisions

	(1)		(2)	
	σ(*Forecast Revision*)		σ(*Forecast Revision*)	
	Coefficient	*t*-statistic	Coefficient	*t*-statistic
Intercept	0.004	18.65 ***	0.003	9.19 ***
Δ*Risk Disclosure (x1,000)*	0.008	1.92 *	0.009	2.37 **
Δ*Non-Risk Disclosure (x1,000)*	-0.001	-1.50	-0.001	-1.71 *
Δ*Market Return Volatility*	-0.000	-0.71	-0.000	-1.06
Δ*Fog Index (x1,000)*			-0.005	-0.34
Δ*Institutional Ownership*			0.001	0.72
Δ*Managerial Forecast*			0.000	0.63
Δ*Sales Growth*			-0.001	-3.37 ***
Δ*ROA*			0.008	3.07 ***
Δ*Segments*			-0.000	-1.68 *
Δ*Loss*			0.004	8.68 ***
Filing Return			0.001	0.53
Absolute Filing Return			0.023	4.94 ***
Δ*Market Return*			0.001	0.98
Number of Analysts			0.000	0.79
N	4,502		4,502	
Adj. R^2	0.1%		5.6%	

This table tests the association between changes in 10-K risk disclosures and changes in the volatility of forecast revisions surrounding 10-K filing dates. Variable descriptions appear in Appendix C. All continuous variables are winsorized at the 1st and 99th percentile, and influential observations with studentized t-statistics greater than two are excluded. Standard errors are heteroskedasticity-adjusted and are clustered for firm and filing month. *, **, and *** denote 10%, 5%, and 1% significance levels, respectively.

Table 5 Firm-level versus industry-level risk disclosures

	(1)		(2)		(3)		(4)		(5)	
	Δσ(*Return*)		Δ(σ(*Neg Ret*)/ σ(*Pos Ret*))		*Log* (*Filing Volume*)		Δ*Log* (*Volume*)		σ(*Forecast Revision*)	
	Coeff.	*t*-stat	Coeff.	*t*-stat	Coeff.	*t*-stat	Coeff.	*t*-stat	Coeff.	*t*-stat
Δ*Firm-Level RD*	0.412	2.08**	0.221	2.34**	0.246	1.66*	0.137	1.20	0.006	1.50
Δ*Industry-Level RD*	3.150	3.32***	1.045	1.97**	1.874	2.03**	2.076	3.86***	0.026	2.35**
Test:Δ*Firm-Level RD* =Δ*Industry-Level RD*		2.89***		1.88*		1.80*		3.63***		1.92*
N	28,110		28,110		28,110		28,110		4,502	
Adj. R^2	2.1%		5.3%		85.3%		7.8%		5.8%	

This table tests the association between changes in the firm-level and industry-level 10-K risk sentences and changes in stock return volatility, trading volume, and volatility of forecast revisions. The control variables are not displayed for brevity. Variable descriptions appear in Appendix C. All continuous variables are winsorized at the 1st and 99th percentile, and influential observations with studentized t-statistics greater than two are excluded. Standard errors are heteroskedasticity-adjusted and are clustered for firm and filing month. *, **, and *** denote 10%, 5%, and 1% significance levels, respectively.